M.I.N.E.R.S. ROVER HARDWARE WIRING GUIDE

Arduino Mega 2560 (Master) + Wemos D1 ESP8266 (Wi-Fi Bridge) | 4-Direction Dual Sonar

CRITICAL COMMON GROUND: Connect all GND lines together on the Breadboard Negative Rail (-): Arduino Mega GND, Wemos D1 GND, all 8 Sonar GNDs, all 3 Gas Sensor GNDs, Left L298N GND, Right L298N GND, and both Battery Negative Black Wires.

1. Power Distribution

• **Logic Power:** USB Powerbank plugs into **Arduino Mega 2560 USB port**. Connect **Mega 5V** to **Breadboard +5V Rail**. Connect **Wemos D1 5V/VIN** to **Breadboard +5V Rail**.
• **Motor Power:** Battery Pack 1 (+7.6V) to **Left L298N VCC**. Battery Pack 2 (+7.6V) to **Right L298N VCC**. Both Battery (-) black wires to **Breadboard GND Rail (-)**.

2. Arduino Mega 2560 <---> Wemos D1 ESP8266 Serial Link

Arduino Mega 2560 Pin	Wemos D1 ESP8266 Pin	Function
TX1 (Pin 18)	RX (GPIO 3)	Mega sends telemetry string to Wemos Wi-Fi bridge
RX1 (Pin 19)	TX (GPIO 1)	Wemos sends laptop commands to Mega
GND	GND	Common Ground Reference

3. Arduino Mega 2560 to Dual L298N Motor Drivers

Remove black jumpers from ENA and ENB on both L298N boards.

Driver / Motor	Arduino Mega 2560 Pin	Function
Left L298N -> IN1	Pin 30	Left Motors Direction 1
Left L298N -> IN2	Pin 31	Left Motors Direction 2
Left L298N -> ENA	Pin 2 (PWM)	Left Motors PWM Speed
Left L298N OUT1 & OUT2	Left-Front + Left-Rear Motors	Parallel Motor Pair Wires
Right L298N -> IN3	Pin 32	Right Motors Direction 1
Right L298N -> IN4	Pin 33	Right Motors Direction 2
Right L298N -> ENB	Pin 3 (PWM)	Right Motors PWM Speed
Right L298N OUT3 & OUT4	Right-Front + Right-Rear Motors	Parallel Motor Pair Wires

4. 4-Direction Dual Ultrasonic Array (2 Sensors per Direction = 8 Total)

All 8 Sonar VCC -> Breadboard +5V Rail | All 8 Sonar GND -> Breadboard GND Rail (-)

Direction Zone	Sensor Unit	Trig Pin (Mega)	Echo Pin (Mega)
FRONT DIRECTION	Front-1 Sensor	Pin 22	Pin 23
FRONT DIRECTION	Front-2 Sensor	Pin 42	Pin 43
RIGHT DIRECTION	Right-1 Sensor	Pin 26	Pin 27
RIGHT DIRECTION	Right-2 Sensor	Pin 46	Pin 47
REAR DIRECTION	Rear-1 Sensor	Pin 50	Pin 51
REAR DIRECTION	Rear-2 Sensor	Pin 48	Pin 49
LEFT DIRECTION	Left-1 Sensor	Pin 24	Pin 25
LEFT DIRECTION	Left-2 Sensor	Pin 40	Pin 41

5. Arduino Mega 2560 to 3x Gas Sensor Array

Sensor Module	Analog Output Pin (AO)	Arduino Mega 2560 Pin	Target Gas Detected
MQ-4	AO Wire	Pin A0	CH4 Methane Gas
MQ-7	AO Wire	Pin A1	Carbon Monoxide (CO)
MQ-2	AO Wire	Pin A2	Smoke / LPG / Flammable Gas